

Optimization Algorithms for MR Reconstruction

Jiaming Liu^a, Rakib Hyder^c, M. Salman Asif^c, and Ulugbek S. Kamilov^{a,b}

^aDepartment of Electrical and Systems Engineering, Washington University in St. Louis, St. Louis, MO, United States

^bDepartment of Computer Science and Engineering, Washington University in St. Louis, St. Louis, MO, United States

^cDepartment of Electrical and Computer Engineering, University of California, Riverside, CA, United States

3.1 Introduction

As discussed in Chapters 1 and 2, the data collection in MRI is performed in the k -space, which corresponds to the Fourier transform of the image. When data is fully sampled, the relationship between the vectorized form of the k -space data $\mathbf{s} \in \mathbb{C}^p$ and the vectorized form of the underlying 2D image $\mathbf{x} \in \mathbb{C}^n$ in MRI can be modeled as

$$\mathbf{s} = \mathbf{F}\mathbf{x}, \quad (3.1)$$

where $\mathbf{F}$ represents the 2D Fourier transform with $F_{kr} = \exp(-i2\pi \mathbf{k}_k^\top \mathbf{r}_r)$, $\mathbf{k}_k \in \mathbb{R}^2$ denotes the location of the k th k -space sample, and $\mathbf{r}_r \in \mathbb{R}^2$ denotes the spatial coordinates of the r th pixel. When both $\{\mathbf{k}_k\}$ and $\{\mathbf{r}_r\}$ are on the Cartesian grid with $n = p$, the k -space measurements $\mathbf{s}$ can be computed using a 2D *fast Fourier transform* (FFT), and the image $\mathbf{x}$ can be easily reconstructed by applying a 2D inverse FFT. In other words, we never have to explicitly store or apply the matrix $\mathbf{F}$. This is true also for the non-Cartesian sampling described in Chapter 4.

Parallel MRI uses a set of coil elements for data acquisition, each of which comes with an independent receiver [1,2]. Each coil element is more sensitive to the magnetization closest to it and less sensitive to magnetization further away, which results in different coil sensitivity profiles in the spatial domain. The fully sampled k -space data recorded by each coil can be represented as

$$\mathbf{s}_c = \mathbf{F}\mathbf{C}_c\mathbf{x}, \quad (3.2)$$

where $\mathbf{C}_c$ is a diagonal matrix that contains the spatial sensitivity map for the c th coil as its diagonal. Note that, even though we represent $\mathbf{C}_c$ as a diagonal matrix, we do not need to store the coil sensitivity maps as a diagonal matrix. Data acquisition with a coil array placed close to the body provides better SNR compared to data acquisition with the body coil, which is integrated in the scanner bore and further away from the measured subject. Parallel imaging refers to applying the coil array for acceleration. This is explained in detail in the chapter on parallel imaging.

As described in Chapter 2, it is desirable in many applications to *accelerate* the MR data acquisition by collecting $p < n$ measurements in the k -space [3,4]. The measurements are typically collected in parallel using $T \geq 1$ receiver coils [1], leading to the model

$$\mathbf{s}_c = \mathbf{B}\mathbf{F}\mathbf{C}_c\mathbf{x} + \boldsymbol{\epsilon}_c,$$

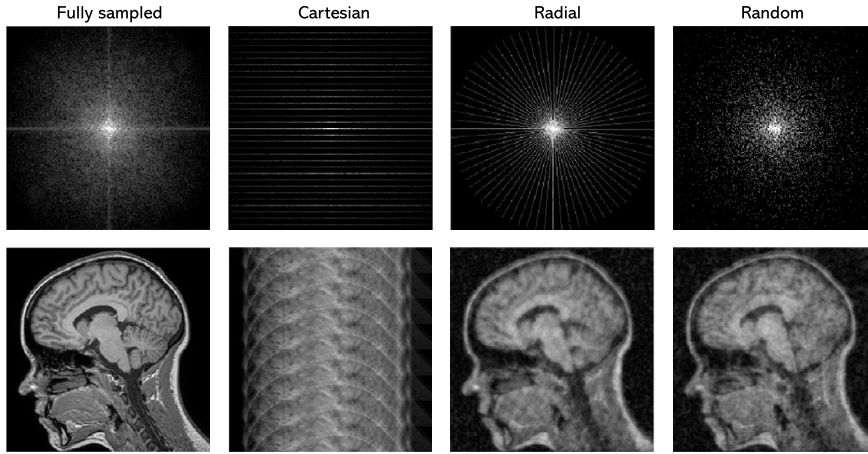


FIGURE 3.1 Sampling strategies in MRI.

Fully sampled k-space can be converted back to the image domain by a simple inverse Fourier transform. *Cartesian*, *radial*, and *random* sampling strategies in this figure correspond to the k-space undersampling by a factor of 8, leading to imaging artifacts in the spatial domain. Note how Cartesian sampling leads to structured artifacts that are more difficult to mitigate computationally.

where $\mathbf{B} \in \mathbb{R}^{p \times n}$ denotes the sampling matrix that contains p rows of the $n \times n$ identity matrix and $\epsilon_c \in \mathbb{C}^p$ is the measurement noise. Note that $\mathbf{B}\mathbf{F}$ does not depend on the coil since all coils see the same k-space sampling pattern. By concatenating the measurements from all coils into the vector $\mathbf{s} = (\mathbf{s}_1, \dots, \mathbf{s}_T)$, we obtain the conventional MRI *forward model*

$$\mathbf{s} = \mathbf{E}\mathbf{x} + \epsilon, \quad (3.3)$$

where $\mathbf{E} \in \mathbb{C}^{m \times n}$, with $m = pT$, is the *measurement operator* representing the physical information and $\epsilon \in \mathbb{C}^m$ is the *measurement noise*, which is often assumed to be additive white Gaussian. The measurement operator $\mathbf{E}$ is assumed to be known during image reconstruction with the sampling matrix $\mathbf{B}$ determined by the location of the k-space samples $\{\mathbf{k}_k\}$ acquired and the sensitivity maps $\{\mathbf{C}_c\}$ determined through a dedicated calibration procedure [5,6]. Note that we do not need $\mathbf{E}$ as an explicit matrix; we can apply $\mathbf{E}$ as a function that computes element-wise multiplication of coil sensitivity maps and image, followed by 2D FFT and subsampling.

The data in MRI is collected using a sequence of measurement trajectories through k-space. The most commonly used trajectories include *Cartesian*, *radial*, and *spiral* (see Fig. 3.1). Additionally, randomized sampling strategies have also been explored in the context of compressive sensing [3,4]. While non-Cartesian sampling generally provides more efficient coverage of k-space and yields less structured artifacts conducive for subsequent image reconstruction, Cartesian sampling is still the most widely-used choice in clinical practice because of its simplicity.

This chapter reviews several widely used optimization strategies for MR image reconstruction. The natural starting point for our discussion is the formulation of MR image reconstruction as an *inverse*

problem, where the acquired MRI data $\mathbf{s}$ is combined and complemented by prior knowledge of the unknown image. Different optimization methods arise naturally based on the properties of the functions used for imposing consistency with the measured data and the image priors. Additionally, modern optimization methods can be conceptually divided into purely model-based reconstruction methods and those that can leverage data-driven priors of unknown images. Throughout the chapter, we seek to explicitly state both the assumptions and results characterizing the convergence properties of discussed iterative optimization algorithms. The readers interested in further exploring the topic of MRI optimization algorithms are encouraged to read a recent review by Fessler [7] that provides a complementary view of the material presented here.

3.2 Least squares reconstruction

Consider the problem of reconstructing an unknown image $\mathbf{x} \in \mathbb{C}^n$ from a set of noisy measurements $\mathbf{s} \in \mathbb{C}^m$ in (3.3). When the matrix $\mathbf{E}$ is well-conditioned with $m \geq n$, the *maximum likelihood (ML)* estimator provides a reasonable approach for reconstructing $\mathbf{x}$. For the *Additive White Gaussian Noise (AWGN)* model, the ML estimator reduces to the *least squares (LS)* problem that has the closed-form solution

$$\hat{\mathbf{x}} = \underset{\mathbf{x}}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{s} - \mathbf{E}\mathbf{x}\|_2^2 = (\mathbf{E}^H \mathbf{E})^{-1} (\mathbf{E}^H \mathbf{s}), \quad (3.4)$$

where $(\cdot)^H$ denotes the Hermitian transpose.

Even though we can represent the solution for the LS problem in closed form, computing the solution can pose challenges in terms of computational and storage complexity if the problem dimension is large. Note that we can apply $\mathbf{E}$ as a computationally efficient operator using element-wise multiplication of vectors and FFTs, but computing $\mathbf{E}^H \mathbf{E} = \sum_{c=1}^T \mathbf{C}_c^H \mathbf{F}^H \mathbf{B}^T \mathbf{B} \mathbf{F} \mathbf{C}_c$ and its inverse is not always efficient.

Suppose we are given fully sampled k-space data from all the receiver coils (i.e., $\mathbf{B}$ is an $n \times n$ identity matrix and $m = Tn$). In such a case, we can simplify $\mathbf{E}^H \mathbf{E} = \sum_{c=1}^T \mathbf{C}_c^H \mathbf{C}_c$ and write the solution in (3.4) as

$$\hat{\mathbf{x}} = \left(\sum_{c=1}^T \mathbf{C}_c^H \mathbf{C}_c \right)^{-1} \sum_{c=1}^T \mathbf{C}_c^H \mathbf{F}^H \mathbf{s}_c, \quad (3.5)$$

where $\sum_{c=1}^T \mathbf{C}_c^H \mathbf{C}_c$ is a diagonal matrix and its inverse can be computed as element-wise inverse of the diagonal entries.

In the case of accelerated MRI, $\mathbf{B}$ is a subsampling matrix with $p < n$, and we cannot simplify the expression for $\mathbf{E}^H \mathbf{E}$ or its inverse. Instead of explicitly computing $\mathbf{E}$, $\mathbf{E}^H \mathbf{E}$ or its inverse, we compute the solution of the LS problem in (3.4) using some iterative method such as the gradient descent or conjugate gradient method [8].

The *gradient descent (GD)* method (which is also called steepest descent) is one of the simplest iterative methods for solving the least squares problem in (3.4). The basic principle in GD method is to iteratively compute the gradient of the LS objective at the current estimate and update the estimate in the negative direction of the gradient with an appropriate step size. Mathematically, we can write the

GD update at iteration t as

$$\mathbf{x}^t = \mathbf{x}^{t-1} - \gamma_t \mathbf{E}^H (\mathbf{E} \mathbf{x}^{t-1} - \mathbf{s}), \quad (3.6)$$

where $\gamma_t > 0$ denotes the step size, which can be adjusted at every iteration or kept fixed. The rate of convergence for GD depends on the choice of step size and the eigenvalues of $\mathbf{E}^H \mathbf{E}$ [9,10].

We can show that the iterative update in (3.6) converges to the LS estimate given in (3.4). Assuming that the gradient descent is initialized with a zero vector $\mathbf{x}^0 = \mathbf{0}$ and $\gamma_t = \gamma$ for all t , we can write

$$\mathbf{x}^t = (I - \gamma \mathbf{E}^H \mathbf{E}) \mathbf{x}^{t-1} + \gamma \mathbf{E}^H \mathbf{s} = \gamma \sum_{k=0}^{t-1} (I - \gamma \mathbf{E}^H \mathbf{E})^k \mathbf{E}^H \mathbf{s}. \quad (3.7)$$

The matrix geometric series $\sum_{k=0}^{t-1} (I - \gamma \mathbf{E}^H \mathbf{E})^k$ converges to $(\gamma \mathbf{E}^H \mathbf{E})^{-1}$ if the spectral norm $\|I - \gamma \mathbf{E}^H \mathbf{E}\| < 1$ [11]. This condition implies that, if

$$0 < \gamma \leq \frac{2}{\lambda_{\max}(\mathbf{E}^H \mathbf{E})}, \quad (3.8)$$

then the gradient descent converges to the LS solution in (3.4), where $\lambda_{\max}(\mathbf{E}^H \mathbf{E})$ represents the largest eigenvalue of $\mathbf{E}^H \mathbf{E}$. Depending on the scaling of $\mathbf{E}$ and the choice of γ , gradient descent may not converge or converge very slowly.

The *conjugate gradient (CG)* method is a popular method for solving a system of linear equations when the system matrix is symmetric and positive definite [12,8]. At every iteration, the CG method updates the estimate in a direction that is *conjugate* to the update directions in all the previous iterations. This property ensures that the CG method will converge in at most n iterations while solving a system with n equations. Suppose we are given a linear system as

$$\mathbf{y} = \mathbf{A}(\mathbf{x}), \quad (3.9)$$

where $\mathbf{A}(\cdot)$ is an operator defined by an $n \times n$ symmetric positive definite matrix. We use the operator notation to highlight that we do not need an explicit matrix form for the operator. A pseudocode with all the steps of the CG method is given in Algorithm 1. Each iteration of CG requires a single application of the system operator $\mathbf{A}(\cdot)$. Note that the LS solution given in (3.4) can be written as a system of linear equations as

$$\underbrace{\mathbf{E}^H \mathbf{s}}_{\mathbf{y}} = \underbrace{\mathbf{E}^H \mathbf{E} \mathbf{x}}_{\mathbf{A}(\mathbf{x})}, \quad (3.10)$$

which can be solved using the CG method given in Algorithm 1. Note that we do not need to compute or apply $\mathbf{E}^H \mathbf{E}$ as an explicit matrix; we can compute the effect of matrix multiplication using a function that involves element-wise multiplication and FFTs.

From the discussion so far, one can consider the CG as a minimizer for the quadratic function, $f(\mathbf{x}) = \|\mathbf{A}(\mathbf{x}) - \mathbf{y}\|^2$. However, the CG can be generalized to minimizing any continuous function, $f(\mathbf{x})$, for which we can calculate the gradient. For the nonlinear CG method, we follow the Algorithm 2. Unlike the linear version of CG, several expressions for calculating β have been developed for nonlinear

Algorithm 1 Conjugate gradient method for solving $\mathbf{y} = \mathbf{A}(\mathbf{x})$

```

1: input: initial estimate  $\mathbf{x}^0$  and residual  $\mathbf{r}_0 = \mathbf{A}(\mathbf{x}^0) - \mathbf{y}$ .
2:  $t = 0$ 
3:  $\mathbf{p}_0 = \mathbf{r}_0$ 
4: repeat
5:    $\alpha_t = \frac{\|\mathbf{r}_t\|_2^2}{\mathbf{p}_t^T \mathbf{A}(\mathbf{p}_t)}$ 
6:    $\mathbf{x}^{t+1} = \mathbf{x}^t + \alpha_t \mathbf{p}_t$ 
7:    $\mathbf{r}_{t+1} = \mathbf{r}_t - \alpha_t \mathbf{A}(\mathbf{p}_t)$ 
8:   if  $\|\mathbf{r}_t\|_2^2 \leq \text{threshold}$  then
9:     break
10:  end if
11:   $\beta_t = \frac{\|\mathbf{r}_{t+1}\|_2^2}{\|\mathbf{r}_t\|_2^2}$ 
12:   $\mathbf{p}_{t+1} = \mathbf{r}_{t+1} + \beta_t \mathbf{p}_t$ 
13:   $t \leftarrow t + 1$ 
14: until convergence/termination criteria satisfied

```

Algorithm 2 Conjugate gradient method for minimizing continuous $f(x)$

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1: input: initial value  $\mathbf{x}^0$ 
2:  $t = 0$ 
3:  $\mathbf{p}_0 = \mathbf{r}_0 = -\nabla f(\mathbf{x}^0)$ .
4: repeat
5:    $\alpha_t = \arg \min_{\alpha_t} f(\mathbf{x}^t + \alpha_t \mathbf{p}_t)$  ▷ via line search
6:    $\mathbf{x}^{t+1} = \mathbf{x}^t + \alpha_t \mathbf{p}_t$ 
7:    $\mathbf{r}_{t+1} = -\nabla f(\mathbf{x}^{t+1})$ 
8:   if  $\|\mathbf{r}_t\|_2^2 \leq \text{threshold}$  then
9:     break
10:  end if
11:   $\beta_{t+1} = \beta_{t+1}^{\text{formula}}$  ▷ Using FR, PR, HS, DY etc formula
12:   $\mathbf{p}_{t+1} = \mathbf{r}_{t+1} + \beta_{t+1} \mathbf{p}_t$ 
13:   $t \leftarrow t + 1$ 
14: until convergence/termination criteria satisfied

```

CG over the years. Four well-known formulas are given by Fletcher–Reeves (FR) [13], Polak–Ribière (PR) [14], Hestenes–Stiefel (HS) [15], and Dai–Yuan (DY) [16]:

$$\beta_{t+1}^{FR} = \frac{\mathbf{r}_{t+1}^T \mathbf{r}_{t+1}}{\mathbf{r}_t^T \mathbf{r}_t}, \beta_{t+1}^{PR} = \frac{\mathbf{r}_{t+1}^T (\mathbf{r}_{t+1} - \mathbf{r}_t)}{\mathbf{r}_t^T \mathbf{r}_t}, \beta_{t+1}^{HS} = \frac{\mathbf{r}_{t+1}^T (\mathbf{r}_{t+1} - \mathbf{r}_t)}{-\mathbf{p}_t^T (\mathbf{r}_{t+1} - \mathbf{r}_t)}, \beta_{t+1}^{DY} = \frac{\mathbf{r}_{t+1}^T \mathbf{r}_{t+1}}{-\mathbf{p}_t^T (\mathbf{r}_{t+1} - \mathbf{r}_t)}.$$

Since the general function f may have many local minima, there is no guarantee that the CG will converge to a global minima. Line search can be used to find α_t in CG as this operation requires zero finding. Two well-known iterative line-search methods for zero-finding are the Newton–Raphson method and the secant method. Both methods require the existence of Hessian matrix of f with respect to α . However, the Newton–Raphson method uses second derivative, whereas the secant method approximates the second derivative using the first derivative. Although the Newton–Raphson method converges much faster, it is computationally expensive.

3.3 Model-based reconstruction

The least squares problem is known to be suboptimal, when $m < n$, because of the ill-posed nature of the inverse problem. We can pose a general recovery problem as the following regularized optimization problem:

$$\hat{\mathbf{x}} = \operatorname{argmin}_{\mathbf{x}} f(\mathbf{x}) \quad \text{with} \quad f(\mathbf{x}) = g(\mathbf{x}) + h(\mathbf{x}), \quad (3.11)$$

where g is the *data-fidelity term* that quantifies the consistency with the observed data $\mathbf{s}$ and h is a *regularizer* that encodes the prior knowledge. The optimization problem in Eq. (3.11) can be recognized as the *Maximum A Posteriori probability (MAP)* estimator of $\mathbf{x}$ for the terms

$$g(\mathbf{x}) = -\log(p_{\mathbf{s}|\mathbf{x}}(\mathbf{s}|\mathbf{x})) \quad \text{and} \quad h(\mathbf{x}) = -\log(p_{\mathbf{x}}(\mathbf{x})),$$

where $p_{\mathbf{s}|\mathbf{x}}$ is the likelihood and $p_{\mathbf{x}}$ is the image prior. Naturally, the LS in Eq. (3.4) is a special case of MAP corresponding to AWGN with a uniform image prior.

In this section, we discuss optimization methods for solving problems of form (3.11). We start by discussing the traditional *smooth optimization* methods applicable to differentiable functions f . We then consider *nonsmooth optimization* methods applicable to problems where the regularizers are not differentiable.

3.3.1 Smooth optimization

Tikhonov regularization is one of the oldest and most well-known techniques for stabilizing ill-posed inverse problems [17, 18]. In its classical form, it seeks to minimize the sum of the LS data-fidelity term and a quadratic regularizer

$$g(\mathbf{x}) = \frac{1}{2} \|\mathbf{s} - \mathbf{E}\mathbf{x}\|_2^2 \quad \text{and} \quad h(\mathbf{x}) = \frac{\lambda}{2} \|\Phi\mathbf{x}\|_2^2,$$

where $\lambda > 0$ denotes a regularization parameter and Φ denotes a sparsifying transform. The Tikhonov solution reduces to the following closed-form expression:

$$\hat{\mathbf{x}} = (\mathbf{E}^H \mathbf{E} + \lambda \Phi^H \Phi)^{-1} (\mathbf{E}^H \mathbf{s}),$$

which can be calculated using the *conjugate gradient (CG)* method in Algorithm 1 [19–21].

In a more general setting, the regularizer h corresponds to a nonquadratic differentiable function. One widely used algorithm for minimizing such functions is the *Accelerated Gradient Method* (AGM) [22,11], whose iterates can be written as

$$\mathbf{x}^t = \mathbf{z}^{t-1} - \gamma_t \nabla f(\mathbf{z}^{t-1}) \quad (3.12a)$$

$$\mathbf{z}^t = \mathbf{x}^t + ((q_{t-1} - 1)/q_t)(\mathbf{x}^t - \mathbf{x}^{t-1}), \quad (3.12b)$$

with $t = 1, 2, 3, \dots$, where $\mathbf{x}^0 = \mathbf{z}^0 \in \mathbb{C}^n$ is the initialization and $\gamma_t > 0$ is the step parameter. Note that, when the *overrelaxation parameters* $q_t = 1$ for all $t \geq 1$, AGM reduces to the traditional gradient descent. However, by adapting $\{q_t\}$ at every iteration, AGM seeks to accelerate the convergence relative to GD. A widely used choice for $\{q_t\}$ is to set it as

$$q_t = \frac{1}{2} \left(1 + \sqrt{1 + 4q_{t-1}^2} \right) \quad \text{starting from } q_0 = 1. \quad (3.13)$$

The step-size parameter of AGM is generally either fixed at a sufficiently small value $\gamma > 0$ or adapted via *backtracking line search* [9]. Backtracking line search seeks to ensure a sufficient decrease in the objective at every iteration. Given the gradient vector $\nabla f(\mathbf{z}^{t-1})$, initial step $\gamma_t > 0$, and parameters $\theta \in (0, 1/2]$ and $\beta \in (0, 1)$, backtracking is performed as follows:

$$\begin{aligned} &\text{while } f(\mathbf{z}^{t-1} - \gamma_t \nabla f(\mathbf{z}^{t-1})) > f(\mathbf{z}^{t-1}) - \theta \gamma_t \|\nabla f(\mathbf{z}^{t-1})\|_2^2 \\ &\quad \text{do } \gamma_t = \beta \gamma_t. \end{aligned}$$

For smooth functions f , this while-loop is guaranteed to terminate for a sufficiently small value of γ_t .

The convergence rate of AGM can be precisely characterized for convex and smooth functions. Consider the following conditions on the optimization problem.

Assumption 1. *The function $f = g + h$ is continuously differentiable and convex. Additionally, the gradient ∇f is Lipschitz continuous with a constant $L > 0$.*

This is a widely-used assumption in the optimization literature that simply asserts the convexity and smoothness of the objective. When f is twice continuously differentiable, Assumption 1 is equivalent to the condition $0 \leq \nabla^2 f(\mathbf{x}) \leq L\mathbf{I}$, where $\nabla^2 f(\mathbf{x}) \in \mathbb{R}^{n \times n}$ is the Hessian matrix of f .

Assumption 2. *The function f has a finite minimum $f^* = f(\mathbf{x}^*)$ attained at some $\mathbf{x}^* \in \mathbb{C}^n$ such that $\|\mathbf{x}^0 - \mathbf{x}^*\|_2 \leq R$.*

The second assumption simply asserts the existence of a minimizer whose distance to the initialization $\mathbf{x}^0$ is bounded. We can now state the following well-known convergence result [11].

Theorem 1. *Run AGM for $T \geq 1$ iterations under Assumptions 1–2 using a fixed step size $0 < \gamma \leq 1/L$. Then, the iterates of the method satisfy*

$$f(\mathbf{x}^{t-1}) - f^* \leq \frac{2R^2}{\gamma(t+1)^2}.$$

Algorithm 3 PGM/APGM

```

1: input: initial values  $\mathbf{x}^0 = \mathbf{z}^0$  and step-size  $\gamma > 0$ .
2: for  $t = 1, 2, \dots$  do
3:    $\mathbf{u}^t \leftarrow \mathbf{z}^{t-1} - \gamma \nabla g(\mathbf{z}^{t-1})$  ▷ more data-consistency
4:    $\mathbf{x}^t \leftarrow \text{prox}_{\gamma h}(\mathbf{u}^t)$  ▷ more regularization
5:    $\mathbf{z}^t \leftarrow \mathbf{x}^t + ((q_{t-1} - 1)/q_t)(\mathbf{x}^t - \mathbf{x}^{t-1})$ 
6: end for

```

As has been shown by Nesterov [11], AGM achieves the optimal $O(1/t^2)$ convergence rate for first-order methods in smooth convex optimization. The *optimized gradient method (OGM)* [23] is a recent alternative to AGM that has a similar per-iteration complexity but leads to a better constant factor LR^2 compared to $2LR^2$ of Theorem 1, achievable using the step parameter $\gamma = 1/L$.

3.3.2 Nonsmooth optimization

Many widely used regularizers $h(\cdot)$ in MRI are nonsmooth. Some well-known examples include the *total variation (TV)* [24,6], *total generalized variation (TGV)* [25,26], and *Hessian Schatten-norm* [27,28]. For example, the anisotropic TV-regularized MRI reconstruction can be formulated using the terms

$$g(\mathbf{x}) = \frac{1}{2} \|\mathbf{s} - \mathbf{E}\mathbf{x}\|_2^2 \quad \text{and} \quad h(\mathbf{x}) = \lambda \|\Phi \mathbf{x}\|_1,$$

where $\lambda > 0$ is the regularization parameter and Φ represents finite differences computed along every dimension of the image $\mathbf{x}$.

Nondifferentiability of widely-used regularizers has led to the widespread adoption of *proximal algorithms* [29] for image reconstruction without differentiation of h . Two widely used algorithms in this context are the *proximal gradient method (PGM)* [30–33] and *alternating direction method of multipliers (ADMM)* [34–36]. The key concept enabling the elegant handling of nonsmooth regularizers is the *proximal operator* [37], defined as

$$\text{prox}_{\lambda h}(\mathbf{z}) = \arg \min_{\mathbf{x} \in \mathbb{C}^n} \left\{ \frac{1}{2} \|\mathbf{x} - \mathbf{z}\|_2^2 + \lambda h(\mathbf{x}) \right\}, \quad \lambda > 0, \quad \mathbf{z} \in \mathbb{C}^n. \quad (3.14)$$

The *projection* onto a subset $\mathcal{X} \subseteq \mathbb{C}^n$ is the proximal operator for $h(\cdot) = \mathbb{1}_{\mathcal{X}}(\cdot)$, where $\mathbb{1}_{\mathcal{X}}(\cdot)$ is the indicator function that takes the value 0 for any $\mathbf{x} \in \mathcal{X}$ and $+\infty$ for $\mathbf{x} \notin \mathcal{X}$. Although the proximal operator can be defined for nonconvex functions, its existence and uniqueness for every $\mathbf{z} \in \mathbb{C}^n$ is guaranteed when h is proper, closed, and convex [38]. The proximal operator can additionally be interpreted as a MAP estimator for the AWGN denoising problem

$$\mathbf{z} = \mathbf{x} + \boldsymbol{\eta}, \quad \text{where} \quad \mathbf{x} \sim p_{\mathbf{x}}, \quad \boldsymbol{\eta} \sim \mathcal{CN}(\mathbf{0}, \lambda \mathbf{I}),$$

by setting $h(\mathbf{x}) = -\log(p_{\mathbf{x}}(\mathbf{x}))$.

Algorithm 3 summarizes both PGM and *accelerated PGM (APGM)* (also known as *Fast Iterative/Shrinkage Thresholding Algorithm (FISTA)*), where the latter is obtained by simply including

Algorithm 4 ADMM

```

1: input: initial value  $\mathbf{x}^0$  and penalty parameter  $\gamma > 0$ .
2: for  $t = 1, 2, 3, \dots$  do
3:    $\mathbf{u}^t \leftarrow \text{prox}_{\gamma g}(\mathbf{x}^{t-1} + \mathbf{z}^{t-1})$  ▷ more data-consistency
4:    $\mathbf{x}^t \leftarrow \text{prox}_{\gamma h}(\mathbf{u}^t - \mathbf{z}^{t-1})$  ▷ more regularization
5:    $\mathbf{z}^t \leftarrow \mathbf{z}^{t-1} + \mathbf{x}^t - \mathbf{u}^t$ 
6: end for

```

the Nesterov acceleration into PGM. In order to implement APGM, one needs to update $\{q_t\}$ as in Eq. (3.13). On the other hand, when $q_t = 1$ for all iterations, the algorithms corresponds to the conventional PGM.

The ADMM-based image reconstruction is obtained by reformulating the inverse problem (3.11) as an equivalent constrained optimization problem. Algorithm 4 summarizes ADMM for the constrained formulation

$$\text{minimize } g(\mathbf{z}) + h(\mathbf{x}) \text{ subject to } \mathbf{z} = \mathbf{x}. \quad (3.15)$$

This constrained problem can then be addressed by forming the *Augmented Lagrangian (AL)* [39] with the dual variable $\mathbf{s}$ and optimizing in an alternating fashion over $\mathbf{x}$, $\mathbf{z}$, and $\mathbf{s}$. One can consider other similar *variable splitting* strategies for obtained different variants of ADMM [40].

Careful inspections of PGM/APGM and ADMM reveal a number of similarities and differences between those two classes of algorithms. Both algorithms alternate between improving data consistency by applying the operators $\mathbf{I} - \gamma \nabla g$ and $\text{prox}_{\gamma g}$, respectively, and imposing the prior knowledge on the image by applying the operator $\text{prox}_{\gamma h}$. The per-iteration complexities of PGM/APGM and ADMM can be dramatically different depending on the difficulty of evaluating corresponding operations. For example, for the LS term, we have

$$\nabla g(\mathbf{x}) = \mathbf{E}^H(\mathbf{E}\mathbf{x} - \mathbf{s}) \quad (3.16)$$

and

$$\text{prox}_{\gamma g}(\mathbf{x}) = \arg \min_{\mathbf{z} \in \mathbb{C}^n} \left\{ \frac{1}{2} \|\mathbf{z} - \mathbf{x}\|_2^2 + \frac{\gamma}{2} \|\mathbf{E}\mathbf{z} - \mathbf{s}\|_2^2 \right\} \quad (3.17a)$$

$$= \left(\mathbf{I} + \gamma \mathbf{E}^H \mathbf{E} \right)^{-1} \left(\mathbf{x} + \gamma \mathbf{E}^H \mathbf{s} \right), \quad (3.17b)$$

which is typically solved with a few iterations of CG.

The convergence behavior of APGM on convex functions is identical to that for AGM. To state it explicitly, we assume the following properties:

Assumption 3. *The function g is continuously differentiable and convex. Additionally, the gradient ∇g is Lipschitz continuous with a constant $L > 0$.*

Assumption 4. *The function h is proper, closed, and convex.*

Assumption 5. *The function $f = g + h$ has a finite minimum $f^* = f(\mathbf{x}^*)$ attained at some $\mathbf{x}^* \in \mathbb{C}^n$ such that $\|\mathbf{x}^0 - \mathbf{x}^*\|_2 \leq R$.*

We can now state the following well-known result in optimization (see proof in [33]).

Theorem 2. *Run APGM for $T \geq 1$ iterations under Assumptions 3–5 using a fixed step size $0 < \gamma \leq 1/L$. Then, the iterates of the method satisfy*

$$f(\mathbf{x}^t) - f^* \leq \frac{2R^2}{\gamma(t+1)^2}.$$

APGM is thus a generalization of AGM that maintains the same optimal worst-case convergence rate. Note that, when the proximal operator corresponds to a projection, APGM reduces to the accelerated variant of the projected gradient method.

The convergence of ADMM has also been extensively investigated in the literature [34–36,40]. In order to state one result, we adopt the following assumptions:

Assumption 6. *The functions g and h are both proper, closed, and convex.*

Assumption 7. *Strong duality holds for the constrained problem in Eq. (3.15).*

Assumption 6 simply states that we are considering a convex optimization problem. A sufficient condition for Assumption 7 is that the interiors of the domains of g and h have common vectors (i.e., their intersection is not empty). We can now state the following classical result (see [40] for a proof).

Theorem 3. *Under Assumptions 6 and 7, the ADMM iterates satisfy the following conditions:*

- Residual convergence: *The iterates approach feasibility*

$$\lim_{t \rightarrow \infty} (\mathbf{x}^t - \mathbf{z}^t) = \mathbf{0}.$$

- Objective convergence: *The objective of the iterates approaches the optimum*

$$\lim_{t \rightarrow \infty} \{g(\mathbf{z}^t) + h(\mathbf{x}^t)\} = f^*.$$

In practice, proximal methods offer faster convergence compared to the traditional techniques based on subgradients or smoothing. Both APGM and ADMM are significantly faster than PGM, which is illustrated on the problem of TV-regularized image reconstruction in MRI in Fig. 3.2. Note the typical behavior of ADMM (discussed in [40]), where it converges to moderate accuracy within a few tens of iterations, but is slow to converge to high accuracy. The rapid convergence of ADMM to moderate accuracy makes it suitable for large-scale image reconstruction, where one is only interested in an approximately good solution.

The primal-dual algorithm can also be used for solving various optimization problems that arise in MRI. A typical primal objective function can be written as

$$\min_{\mathbf{x}} g(\mathbf{E}\mathbf{x}) + h(\mathbf{x}) \quad (\text{primal}), \quad (3.18)$$

where $g(\mathbf{E}\mathbf{x})$ is the data fidelity and $h(\mathbf{x})$ is the regularization. The Chambolle–Pock (CP) algorithm, as proposed in [41], is a first-order primal-dual method that solves the following saddle-point problem:

$$\min_{\mathbf{x}} \max_{\mathbf{v}} \langle \mathbf{E}\mathbf{x}, \mathbf{v} \rangle + h(\mathbf{x}) - g^*(\mathbf{v}) \quad (\text{primal-dual}), \quad (3.19)$$

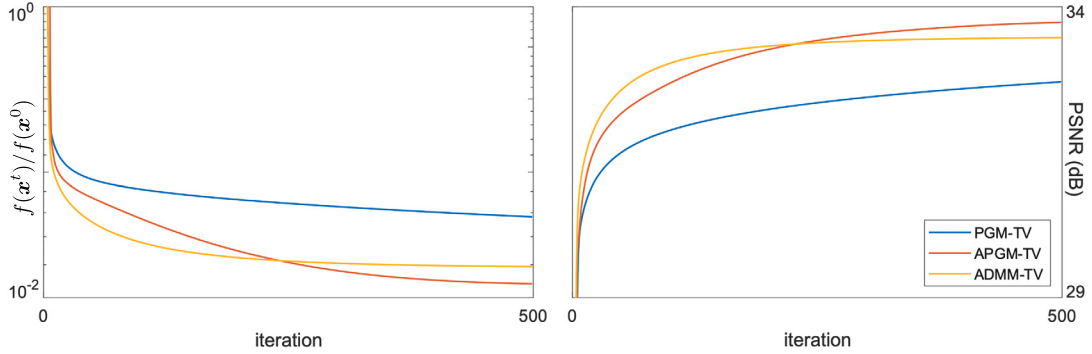


FIGURE 3.2 Convergence behavior of PGM, APM, and ADMM.

The evolution of the relative objective (left) and PSNR (dB) (right) for MR image reconstruction with the TV regularizer. Both APM and ADMM are known to lead to faster convergence compared to the traditional PGM.

Algorithm 5 Primal-dual method in [41,42]

- 1: **input:** initial value $\mathbf{x}^0, \bar{\mathbf{x}}^0, \mathbf{v}^0$
 - 2: $t = 0$
 - 3: Choose step size $\sigma > 0$ and $\tau > 0$, so that $\sigma\tau L^2 < 1$, where $L = \|\mathbf{E}\|$ is the induced norm, and $\theta \in [0, 1]$.
 - 4: **repeat**
 - 5: $\mathbf{v}^{t+1} = \text{prox}_{\sigma g^*}(\mathbf{v}^t + \sigma \mathbf{E} \bar{\mathbf{x}}^t)$ ▷ dual proximal
 - 6: $\mathbf{x}^{t+1} = \text{prox}_{\tau h}(\mathbf{x}^t - \tau \mathbf{E}^* \mathbf{v}^{t+1})$ ▷ primal proximal
 - 7: $\bar{\mathbf{x}}^{t+1} = \mathbf{x}^{t+1} + \theta(\mathbf{x}^{t+1} - \mathbf{x}^t)$ ▷ extrapolation
 - 8: $t \leftarrow t + 1$
 - 9: **until** convergence/termination criteria satisfied
-

where $g^*(\mathbf{x})$ is the convex conjugate of $g(\mathbf{x})$ [9]. The reasoning behind moving towards the primal-dual approach instead of solving the primal function lies directly in the nontrivial formulation of the proximal operator for $g(\mathbf{E}\mathbf{x})$. We can formulate proximal operator for $g^*(\mathbf{x})$ and $h(\mathbf{x})$ easily. We alternately do proximal gradient descent for $\mathbf{x}$ and $\mathbf{v}$ to solve the optimization in (3.19) according to Algorithm 5. Over the years, many variations of the primal-dual approach have been proposed to solve smooth and nonsmooth optimization [41–48].

$g(\mathbf{E}\mathbf{x})$ can be a nonsmooth function, such as hinge loss [49], generalized hinge loss [50], absolute loss [51], or ϵ -insensitive loss [52]. $h(\mathbf{x})$ can be a nonsmooth regularizer, such as lasso [53], group lasso [54], sparse group lasso [55], exclusive lasso [56], $l_{1,\infty}$ regularizer [57], or trace-norm regularizer [58].

3.3.3 Stochastic gradient-based approaches

Stochastic versions of gradient descent methods have become popular with the rise of neural networks. Stochastic gradient descent (SGD) methods have been extremely successful in solving various non-

convex optimization problems in recent years. Multiple variations of the GD and SGD methods have been proposed over the years to ensure *better* convergence. Such approaches include Nesterov accelerated gradient (NAG) [59], Adagrad [60], RMSprop [61], Adadelta [62], adaptive moment estimation (ADAM) [63], and Nesterov-accelerated adaptive moment estimation (NADAM) [64]. Nesterov accelerated gradient (NAG) incorporates the momentum term to the GD update steps. As the GD iterations approach a local minima, the updates usually become gradually smaller, otherwise we will jump from the vicinity of one local minima to another. To ensure that, the momentum term is introduced. The momentum term increases for parameters whose gradients point in the same directions and reduces updates for parameters whose gradients change directions. Adagrad adapts the learning rate of the parameters based on the frequencies of the features. Adagrad has diminishing learning rate for infrequent features. RMSprop and Adadelta have both been developed to resolve Adagrad's diminishing learning rate. Adam uses the first and second moments of the gradients for GD updates. NADAM incorporates Adam and NAG by modifying the momentum term.

3.4 Summary

This chapter reviewed several widely used optimization strategies in the context of MRI image reconstruction. The key property desired in such algorithms is the ability to leverage useful prior information about the unknown image $\mathbf{x}$. We have discussed algorithms for handling smooth and nonsmooth regularizers.

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